Analyzing data with Python

Python Programming for Business Intelligence

Sippo Rossi
April 22nd 2026

Learning goals

Theoretical:

- Understand when to use different types of visualizations
- Understand what libraries are available for statistical analysis
- Understand the principles of text analysis with LLMs

Practical:

- Know how to use seaborn
- Know how to run an A/B test using Python
- Know how to do basic text analysis using an LLM

Motivation

Once you have finished collecting and cleaning the data, you can move on to analyzing it. In this lecture, you will encounter three different types of data analysis: visual, statistical and text

Each of these topics could be a full course on its own and it is not expected that you will master any of the topics based on this lecture alone

The purpose of this session is to give you examples and motivation to further study these topics independently, and possibly apply them in the project, based on your own interests

The business intelligence process visualized

Recall this slide from last week

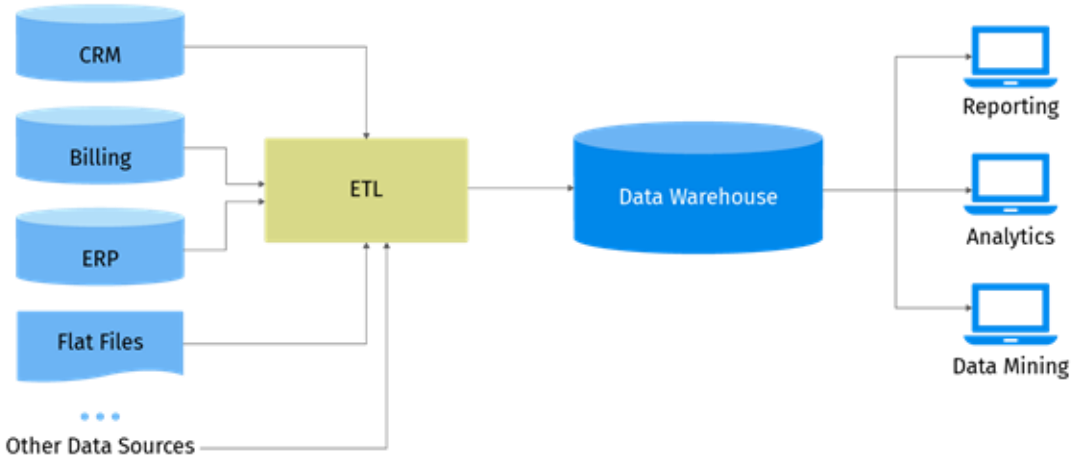


Image from: https://blog.griddynamics.com/edw-performance-comparison/

The business intelligence process visualized

Now we are moving onto the final part of the process

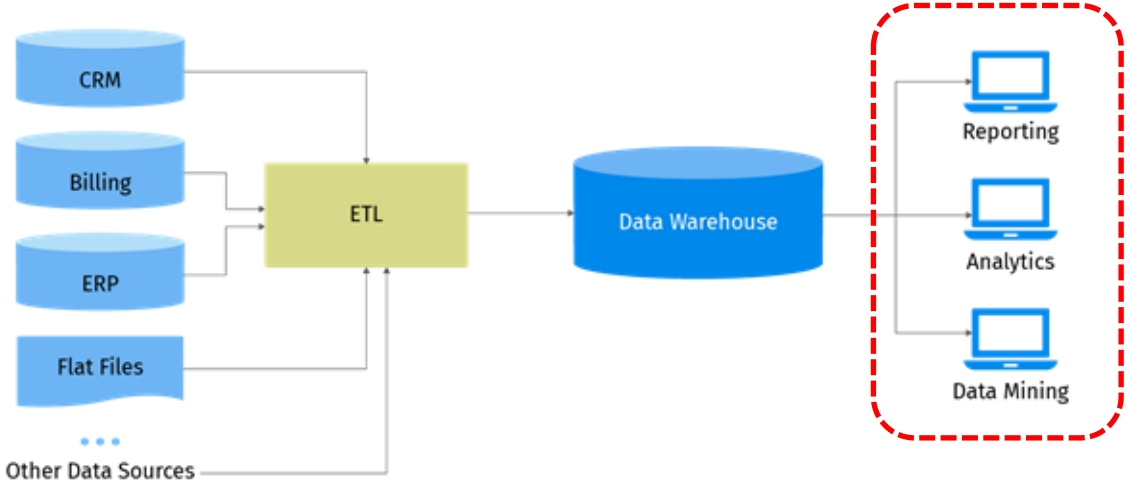


Image from: https://blog.griddynamics.com/edw-performance-comparison/

Agenda

- Data visualization
- Statistical analysis:
    - Libraries
    - Descriptive statistics
    - A/B testing
- Text analysis

Data visualization

The role of data visualizations

Last week covered how visualizations can be used to understand the data during EDA

Beyond EDA, visualizations are central to analysis, reporting, and communicating results

Visualization-driven analysis (visual analytics) has become a core part of BI in many organizations

Different types of visualizations

During this lecture we will cover the most common and generally usable types of visualizations

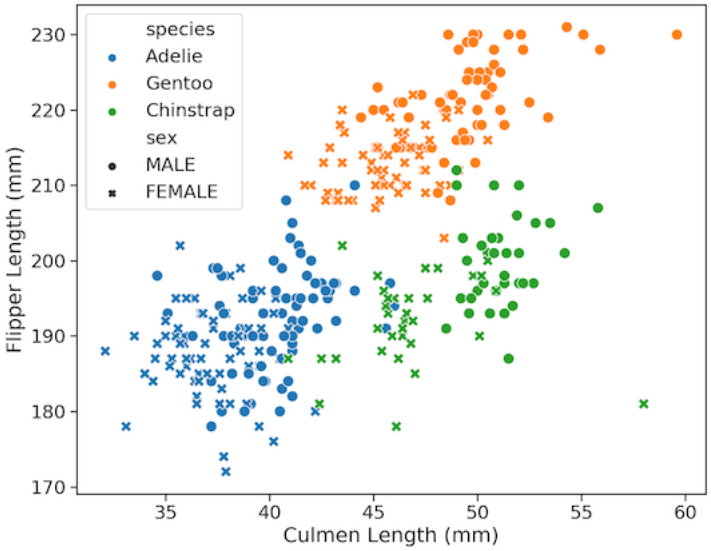
While seaborn supports many forms of visualizations, I encourage you to always think whether the same information would be better presented using one of the following instead of opting for more exotic plots:

- a bar chart
- a histogram
- a scatter plot
- a line chart
- a heatmap
- a boxplot

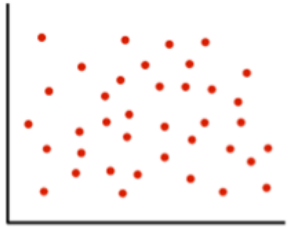
Scatter plots

Scatter plots are useful for inspecting bivariate data or pairs of variables

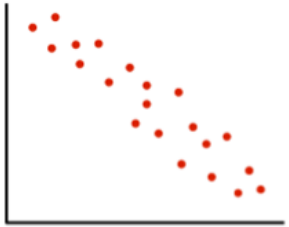
Scatter plots reveal e.g., clusters or if there is a positive, negative or no correlation at all between variables



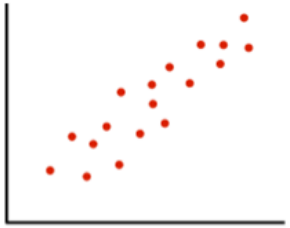
What type of a correlation is this?



No Correlation



Negative



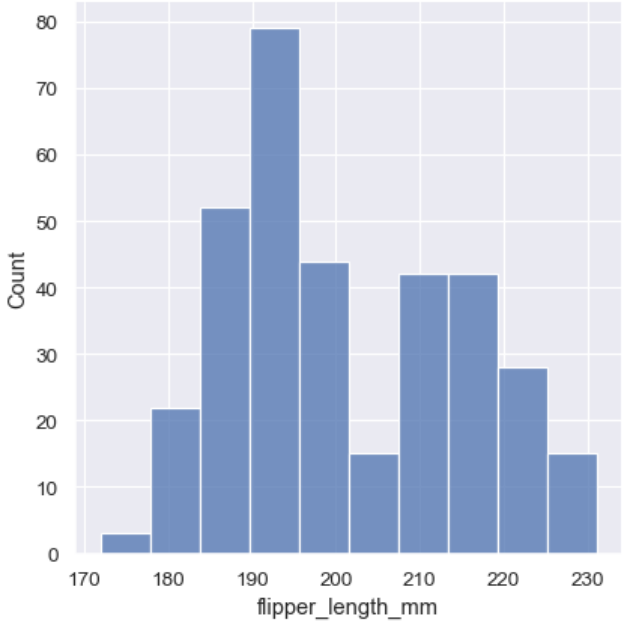
Positive

Histograms

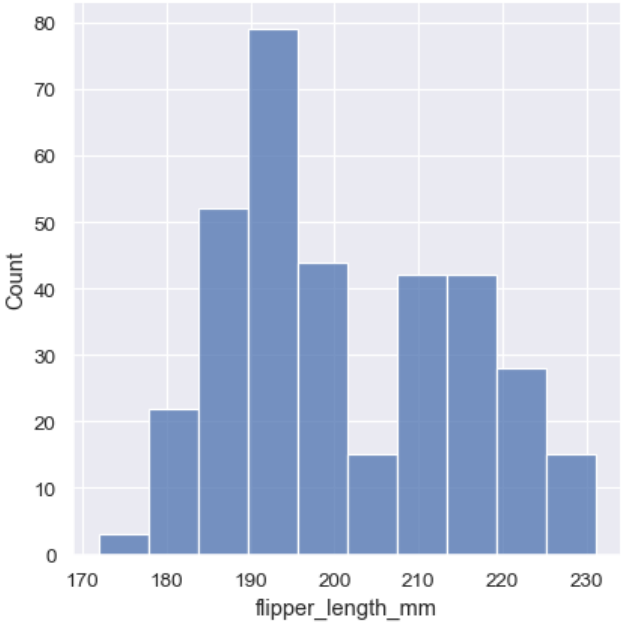
Histograms can help understand how variables are distributed

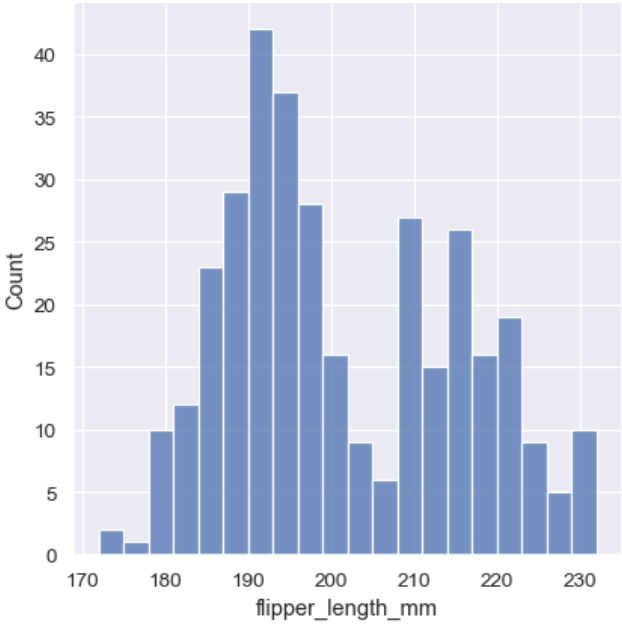
The main parameter is bin size which determines how the data is divided

The size of the bins can significantly affect the interpretation of the plot



Same data but different bin size

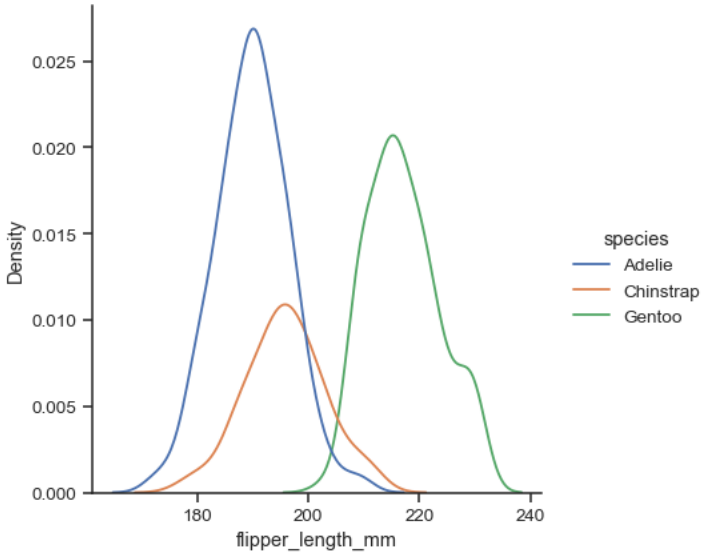




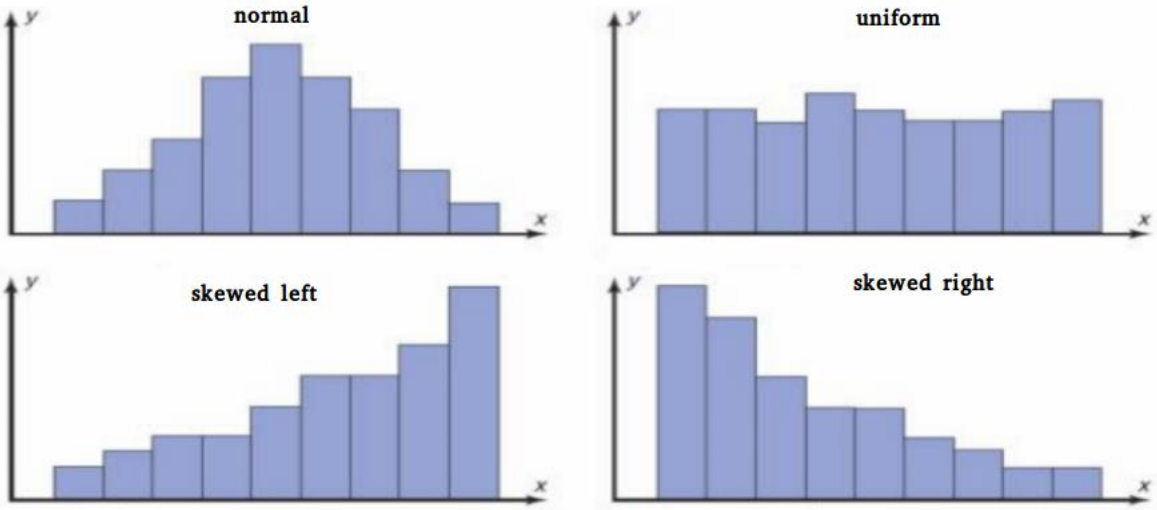
Different types of ways to plot distributions

With seaborn there are several ways to plot distributions using e.g. the displot or kdeplot functions

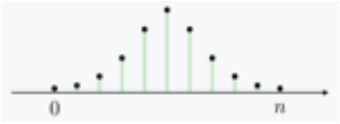
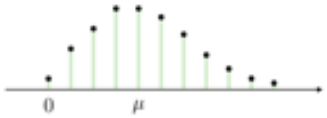
Density plots are slightly more difficult to interpret than histograms but do not have the issues related to selecting bin size



Some examples of different distributions

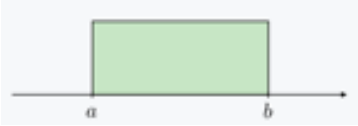


Discrete distributions

Distribution	Notation	Use	Illustration
Discrete uniform distribution	X~U(a,b)	When the experiment has a known, finite number of outcomes equally likely to happen such as throwing dice	
Binomial distribution	X~B(n,p)	When interested in the number of successes in a sequence of n independent experiments	
Poisson distribution	X~Po(λ)	When the probability of a given number of events occurring in a fixed interval of time or space and if these events occur with a known constant mean rate and independently of the time since the last event	

Images by Shervine and Afshine Amidi from https://stanford.edu/~shervine/teaching/cme-106/cheatsheet-probability

Continuous distributions

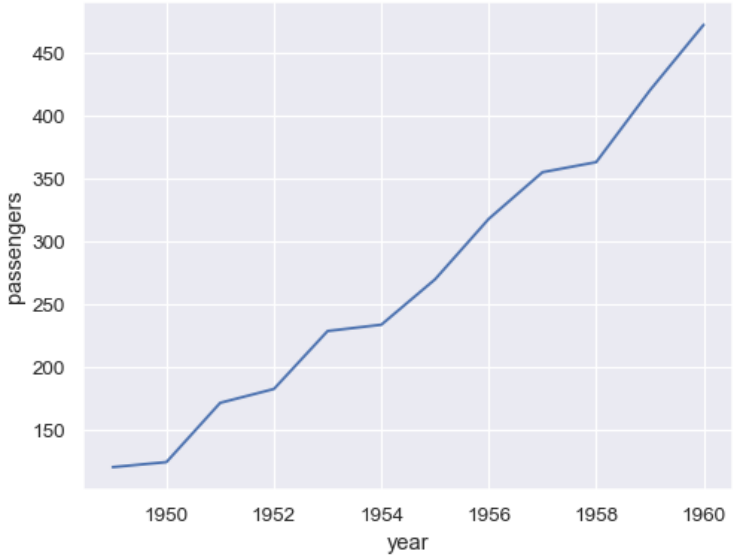
Distribution	Notation	Use	Illustration
Uniform distribution	X~U(a,b)	When in an experiment there is an arbitrary outcome that lies between certain bounds	
Gaussian distribution (Normal distribution)	X~N(μ,σ)	The most widely used distribution as it occurs commonly in nature and in other phenomenon	
Exponential distribution	X~Exp(λ)	When events that occur continuously and independently at a constant average rate	

Images by Shervine and Afshine Amidi from https://stanford.edu/~shervine/teaching/cme-106/cheatsheet-probability

Line charts

Line charts are a good way to visualize time series data

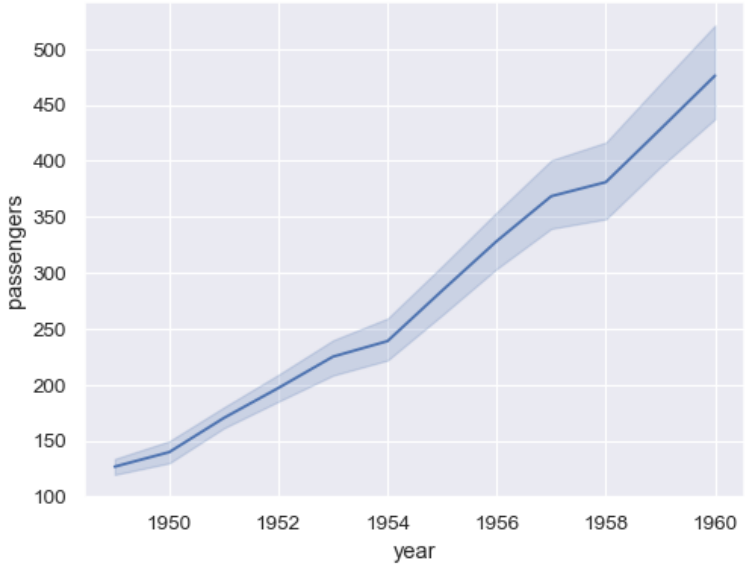
With this visualization it is easy to see trends



Line charts and confidence intervals

Some seaborn functions can plot statistical information

If you pass to a Lineplot data in long-form and there are multiple y values for the same x value, it will plot the 95% confidence interval and mean

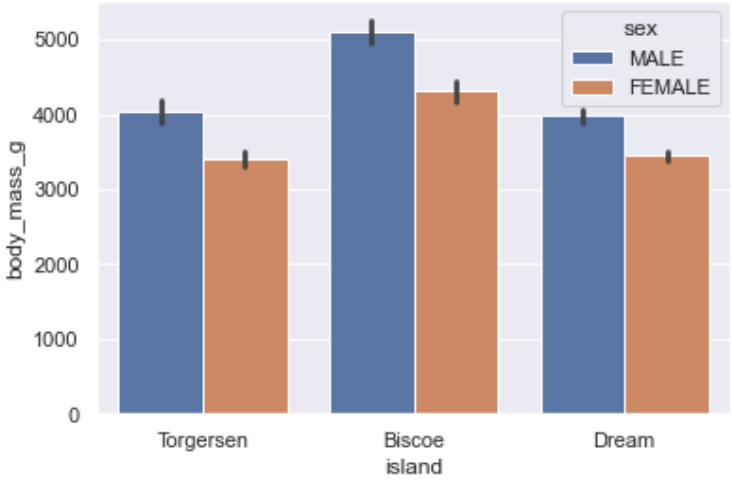


Bar charts

Bar charts are useful for presenting categorical data where one axis is the category and other a measured value

The example on the right shows for three different penguin species the average weight of female and male specimens

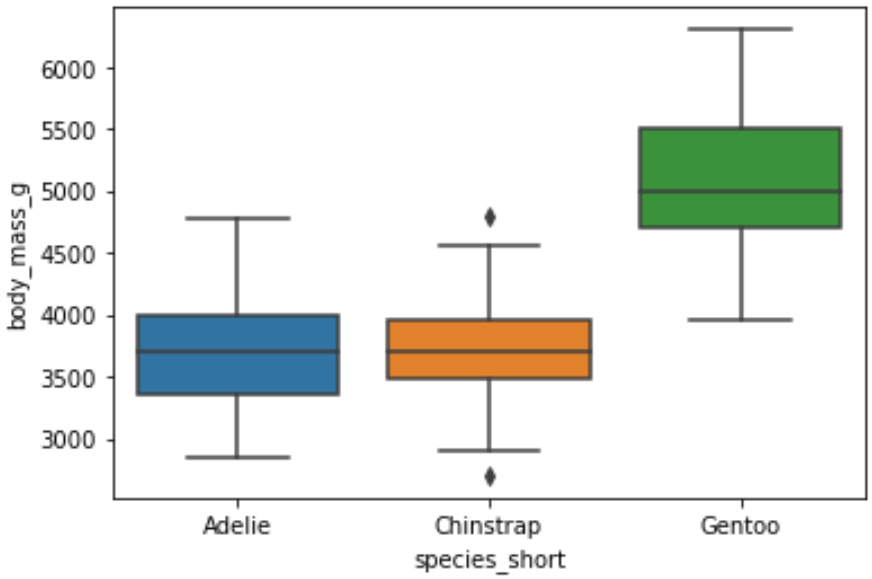
Bar charts should not be confused with histograms



Boxplots

Box plots shows the three quartile values of the distribution as well as the range

Box plots are useful for comparing distributions in categorical data

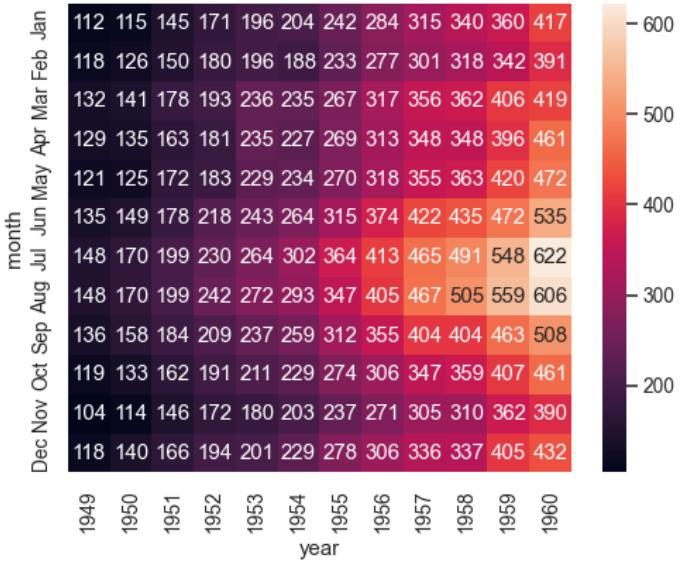


Heatmaps

Heatmaps are an easy way to see relationships between two variables and their different values

The color describes the magnitude of some measurement

In the example it is the number of flights by month and year with smaller numbers being darker



Other Python libraries for visualizations

Bokeh can be used to create html pages viewable with a browser with interactive visualizations:

https://bokeh.pydata.org/en/latest/

Plotly can also be used to create interactive visualizations with Python:

https://plotly.com/python/

seaborn demo

Going through *L5_1_seaborn.ipynb*

Statistical analysis

Statistical analysis with Python

There are several widely used Python modules/libraries for statistical analysis:

- statistics, Python's built in module, which covers basic descriptive statistics, although typically this is handled with pandas
- scipy.stats, SciPy's statistics module, which has a lot of tools for probability distributions and hypothesis testing. Works on array-like data (e.g. NumPy arrays / pandas series)
- statsmodels, which uses R-style formulas and table outputs and works on pandas DataFrames. Based on creating a "fitted" model object
- PyMC and PyStan, Python libraries for Bayesian statistics

Which module / library should be used?

It depends

Different projects and workflows require different tooling

There are also many niche libraries that focus on some specific statistical analysis task, e.g. Orbit, a library for Bayesian time series forecasting and inference

Statistical analysis in this course

There are many other courses at Hanken which are better for learning about statistics. Thus, statistics is intentionally only covered briefly in this course

In this course we will focus on two examples:

- Calculating descriptive statistics (covered last week)
- A/B testing

Descriptive statistics

In addition to being used during EDA, summary statistics (mean, median, mode, variance, standard deviation and range) are often reported to give an overview of the data

Other descriptive statistics, such as the distribution of data within specific variables is often also included

Examples of this include how in publications that contain human data, we often report demographics with a table

Table 1. Demographic data of the survey respondents (*n* = 311)

Characteristic	No.	%
Sex		
Male	141	45.3
Female	170	54.7
Age (years)		
≤ 20	66	21.2
21–40	112	36.0
41–60	123	39.0
≥ 61	10	3.2
Education level		
Primary school	31	9.9
Vocational or secondary school	102	32.8
Higher or tertiary education	99	31.8
Master's degree/doctorate	51	16.4

A/B testing

A/B testing is a widely used method for supporting business decisions through statistical hypothesis testing

The basic goal is to run a randomized controlled experiment with two (or more) variants, A and B

For example, showing users different versions of a website and measuring performance. If version B results in 20% more sales than version A, statistical analysis can determine whether this difference is statistically significant. If so, all customers could be shown version B going forward

A/B testing marketing email example

Group A: Gets an email with the subject "50% Off This Weekend!”

Group B: Gets an email with the subject "Special Deal Just for You!”

If more people open the email from Group B and the result is statistically significant, you can decide that the second subject line is better.

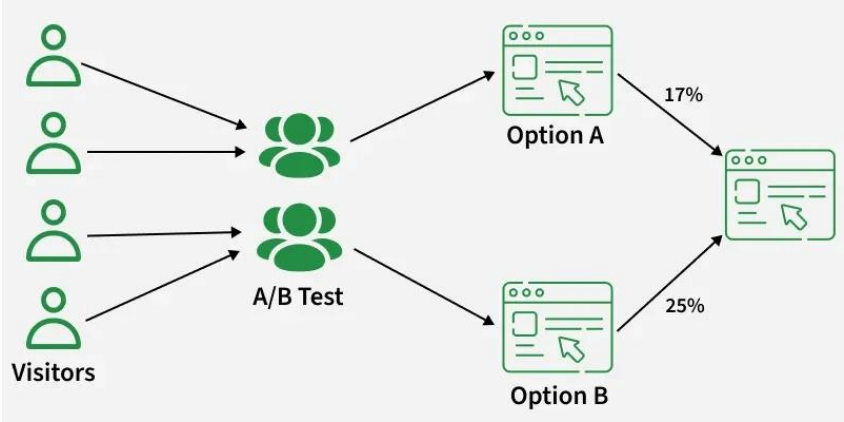


Image and example from: https://www.geeksforgeeks.org/data-science/a-b-testing-using-python/

A/B testing demo

Going through *L5_2_ab_testing.ipynb*

Text analysis

Natural language processing and text analysis

As previously stated, natural language processing (NLP) and text analysis could be a full course on its own

We will briefly explore basic text analysis tasks such as sentiment analysis and text classification using an LLM

The purpose of this topic is to give you some experience working with NLP-related tasks and to provide more practice with using APIs

Text analysis with LLMs via an API

LLM APIs make text analysis accessible without needing to build processing pipelines and custom NLP models, but tokens cost and expenses can add up

LLMs are probabilistic, not deterministic. Thus, they are best suited for low-stakes tasks (e.g., categorizing customer feedback, summarizing social media comments) where occasional errors are acceptable

Examples of use cases include sentiment analysis of product reviews, classifying support tickets by topic, extracting key themes and main points from open-ended survey responses

Text analysis with local LLMs

Tools such as Ollama allow you to run open source LLMs locally (the model running on your computer), eliminating API costs and keeping data on-premises

However, they requires capable hardware (sufficient RAM and ideally a GPU), and local models are typically smaller and much less capable than closed frontier models such as GPT-5.4 or Claude Opus 4.6

Sentiment analysis

Sentiment analysis aims to determine the emotional tone or opinion expressed in a piece of text (e.g., positive, negative, neutral)

With an LLM, you can perform sentiment analysis by prompting the model to classify the sentiment of a given text. With a Python script that uses an LLM through an API, this can be applied to an entire column of data

Examples of use cases include analyzing product reviews, monitoring brand perception on social media posts, gauging employee satisfaction from survey responses

Text classification

Text classification assigns predefined categories or labels to a piece of text e.g., assign a topic from these options: technology, healthcare, finance, energy, other

Sentiment analysis is technically a specific case of text classification, but the term usually refers to broader categorization tasks

With an LLM, you define the categories in the prompt, and the model classifies each text accordingly

Examples of use cases include classifying emails to customer service based on severity and tagging news articles by topic

Text analysis demo

Going through *L5_3_text_analysis.ipynb*

Conclusion

Conclusion

This week we covered three very different forms of data analysis

During the last two weeks the focus will be on machine learning

Remember to submit the peer review today before midnight

Questions, comments suggestions?